Introduction to quantitative finance

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Outline

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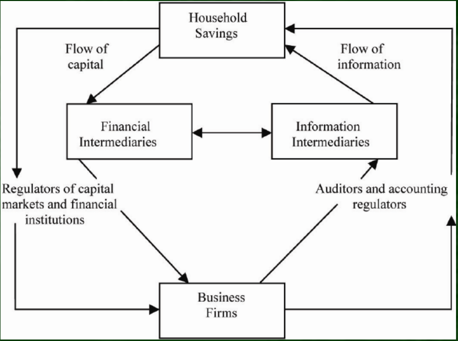
   - Markowitz's portfolio diversification
   - CAPM theory
   - Arbitrage pricing theory
4. Term structure models for interest rates
5. Volatility modelling
   - Classical models
   - Open problems
6. Risk measures

References

- A. Shiryaev. Essentials of stochastic finance: Facts, Models, Theory.
- P. Wilmott. Introduces Quantitative Finance.
- J-A. Yan. Introduction to Stochastic Finance.
- Many other sources

Objects and structures

- Individuals
- Corporations
- Intermediaries
- Financial markets



Financial markets: Money

- Means of payment: Intermediate method to trade "things one has" for "things/services one wants/needs to get". Evolve to Transactions.
- Types of currency from history: sea-shelves, metal coins, notes, crypto-currencies.
- Storage of wealth: measure of value, saving means.
- The privilege of issuing money is synonymous with economic power.

Financial markets: Foreign currencies & exchange rates

Percent Change in the Last 24 Hours			
EUR/USD	-0.72321%	USD/JPY	+0.07883%
GBP/USD	-0.16806%	USD/CHF	+0.33184%
USD/CAD	+0.98822%	EUR/JPY	-0.64495%
AUD/USD	-1.43610%	CNY/USD	-0.13479%

- Gold standard of money: US dollar can be exchanged to gold coin and back.
- Pool of currencies: means of payment in foreign trade. USD, EUR, JPY, GBP, CHF, RMB, VND. Monetary Union (IMF).
- Bretton Woods credit and currency system (1944): foreign exchange rates between currencies. Fixed cross rates ($\pm 1\%$ band).
- 1973 financial crisis: break of Bretton-Woods system. Replaced by floating exchange rate.
- 1979: European Monetary System with $\pm 2.25\%$ to keep the exchange rates stable. Later Euro currency (EUR).

Financial markets: Precious metals

London Vault — toz kg $ £ €

Gold	Silver	Platinum	Palladium
Buy	Buy	Buy	Buy
$1,848	$21.71	$979	$1,951
Sell	Sell	Sell	Sell
$1,843	$21.65	$975	$1,911

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- Gold, silver, platinum, iridium, ... and other precious metals.
- Means of value/wealth storage. Jewelry (like seashelves from past times). Some brainstorming ideas?
- Gold standard periods: 1821 with British pound/sterling, followed by US. Blossomed in 1880-1914. After WWI deteriorated. Disappeared in 1971 after US financial crisis.
- Today gold is still reserved by central banks and price is still on the long term way up, due to inflation and credit growth.

Financial markets: Bank accounts

- Interest saving rates and loan rates: banks borrow money from saving account holders and loan it to business firms. Loan rate is usually higher than borrow rate, resulting in a surplus/profit in banking business.
- Simple and Compound rates: an account of capital B_0 paying m times a year with interest rate $r(m)$ will obtain in N years an amount

$$B_N(m) = B_0(1 + \frac{r(m)}{m})^{mN} \to B_N(\infty) = B_0 e^{r(\infty)N} \tag{1}$$

as $r(m) \to r(\infty), m \to \infty$. In practice, one can roughly estimate

$$r(m) = m(e^{\frac{r(\infty)}{m}} - 1) \text{ and } r(\infty) = m\ln(1 + \frac{r(m)}{m}). \tag{2}$$

Financial markets: Bonds

- Definition: promissory notes issued by a party (governments, banks, corporations...) to raise capital. Written as Debt in the balance sheet.
- Interest rate is fixed and payable regularly. The repayment of the entire loan at the expired date is guaranteed, but not 100 % sure (go broke or default). Government bonds are less risky than corporate bonds, but the interest rate is lower (implication of no arbitrage theory).
- T: maturity date; $P(0, T)$: issued/original bond price; $P(T, T)$: face value; r_c (fixed): bond's interest rate (coupon yield) defining dividends. $P(t, T)$ varies due to many economic factors: supply and demand, bank interest rates, speculators' activities...
- Current yield: $r_c(t, T) = \frac{r_c P(T,T)}{P(t,T)}$. Yield to maturity (profitability) $\rho = \rho(T - t, T)$ is the discounted rate that solves the equation

$$P(t, T) = \sum_{k=1}^{T-t} \frac{r_c P(T, T)}{(1+\rho)^k} + \frac{P(T, T)}{(1+\rho)^{T-t}}. \quad (3)$$

Financial markets: Shares

- Definition: notes issued by banks, corporations.. to raise capital. Written as Equity in the corporation's financial balance sheet.
- Types: ordinary shares and preferred shares, which differ in riskiness and dividend payments.
- When a corporation/bank goes bankrupt, bond holders receives the repayment first, then the preferred share holders, the rest comes to ordinary share holders.
- Prices of bonds and shares fluctuate by corporation's (quarterly/yearly) earning reports, dividend payments, and the expectation on the future business activities.
- Long term investors are interested in fundamental factors: business cycles, corporate ranking and revenue/profit growth. Speculators and traders (chartists) take advantage of share price's fluctuation.
- Corporation value pricing methods: price/earning ratio, DCF...

Trading psychology

- In the long term, stock prices reflect the fundamental values of a corporation. But in the short run, prices are influenced by traders' psychology, opinions, rumors and ... weather!!!
- R. J. Shiller (Nobel 2013): Irrational exuberance refers to investor enthusiasm that drives asset prices higher than those assets' fundamentals justify.
- Over optimistic psychology leads to market bubbles, while too pessimistic sentiments drive the panic selling.
- Bulls make money, bears make money, but pigs get slaughtered.

Traders' psychology

Market trading activities

International Herald Tribune, October 27, 1989. Kal, Cartoonists and Writers Syndicate, 1989.

Derivative market: Overview

- One needs to develop financial technology to match the demands from parties for hedging products in businesses. It looks like buying an insurance in case the unexpected event happens.
- It is called derivatives, because it is based on the price of a primary asset/product, which can be commodities or shares.
- 1973: first exchange was opened to trade option contracts: Chicago Board Option Exchange (CBOE). Trading volume increases from 900 contracts in 1973 to 20k in 1974 to 100k in 1976, and to 700k contracts in 1987. Today the volume is 40 M contracts.
- 1973: two papers on option pricing methods: "The pricing of options and corporate liabilities" (F. Black and M. Scholes), and "The theory of rational option pricing" (R.C. Merton).

Derivative products

- Call/Put Option: a security (contract) issued by a firm, bank, financial institution and giving the buyer the right to buy/sell something of value (a share, bond, currency,..) on specified terms at a fixed instant or during a certain period of time in the future.
- Future contract: a commitment to buy/sell something of value (gold, cereals, foreign currency) at a present instant of time in the future at a (future) price fixed at the moment of signing the deal.
- Forward contract: an agreement to deliver/buy something in the future at a specified (forward) price.
- Future vs. Forward: Forward is an agreement between two parties only, while with Future there is another third party involved as intermediaries, like specialized exchange.
- Terminology: Buy = holding a Long position, Sell = holding a Short position.

Market assumptions for modeling

- Filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F})$ where $\mathbb{F} = (\mathcal{F}_n)$ is an information flow.
- Random walk conjecture:
- Complete market hypothesis: negligible transaction costs and therefore also perfect information; plus there is a price for every asset in every possible state.
- Efficient market hypothesis: $\frac{S_t}{B_t} = \mathbb{E}_{\hat{\mathbb{P}}}(\frac{S_{t+1}}{B_{t+1}}|\mathcal{F}_t)$ with a risk neutral probability measure $\hat{\mathbb{P}}$ that is equivalent to $\mathbb{P}$, i.e. $(\frac{S}{B})$ is a $\hat{\mathbb{P}}$-martingale.
- Example: Binomial and Brownian motion models of stock prices:

Data, time scale

- Time series of data: $S_{\tau_1}, S_{\tau_2}, \dots S_{\tau_n}, \dots$, where τ_k can be tick, minutes, hours, daily, weekly, monthly, yearly. If τ_k is not regularly updated, then one needs to use continuous modification $(\tilde{S}_t)$ where

$$\tilde{S}_t = S_{\tau_k} \frac{\tau_{k+1} - t}{\tau_{k+1} - \tau_k} + S_{\tau_{k+1}} \frac{t - \tau_k}{\tau_{k+1} - \tau_k}, \quad \forall t \in [\tau_k, \tau_{k+1}]. \tag{4}$$

- Interested in log-return $r_{\tau_k} = \ln \frac{S_{\tau_{k+1}}}{S_{\tau_k}}$ and its modification $\tilde{r}_{t_k} = \ln \frac{\tilde{S}_{t_{k+1}}}{\tilde{S}_{t_k}}$. Assumption: $\tilde{S}_{t_k}$ are random variables, and so are $\tilde{r}_{t_k}$. Furthermore, $\tilde{r}_{t_k}$ are identically distributed, i.e. $(\tilde{S}_{t_k})_k$ is a stochastic process with stationary increments.
- Average $\bar{r}_N = \frac{1}{N}\sum_{k=1}^N \tilde{r}_{t_k}$. Moments: $\hat{m}_p = \frac{1}{N}\sum_{k=1}^N (\tilde{r}_{t_k} - \bar{r}_N)^p$.
- Empirical skewness $\hat{S}_N = \frac{\hat{m}_3}{(\hat{m}_2)^{\frac{3}{2}}}$. Empirical kurtosis $\hat{K}_N = \frac{\hat{m}_4}{(\hat{m}_2)^2} - 3$.

Distributions

- Not Gaussian but heavy tails: $\mathbb{P}(\tilde{r}_{t_k} > x) \approx x^{-\alpha} L(x)$ where $\alpha > 0$, L-slowly varying, $\frac{L(xy)}{L(x)} \to 1$ as $x \to \infty$ for each $y > 0$. Suggested Student's t-distribution: $f_n(x) = \frac{1}{\sqrt{\pi n}} \frac{\Gamma(\frac{n+1}{2})}{\Gamma(\frac{n}{2})}(1 + \frac{x^2}{n})^{-\frac{n+1}{2}}$.
- Stable distribution (Mandelbrot, Fama): Pareto distribution

$$f_{\alpha b}(x) = \begin{cases} \frac{\alpha b^\alpha}{x^{\alpha+1}} & x \geq b, \\ 0 & x < b. \end{cases} \tag{5}$$

Maximum likelihood estimator $\hat{\alpha}_N$ on N independent observations $(X_1, \ldots, X_N)$ satisfies

$$\max_\alpha \prod_{k=1}^{N} f_{\alpha b}(X_k) = \prod_{k=1}^{N} f_{\hat{\alpha}_N b}(X_k) \Rightarrow \frac{1}{\hat{\alpha}_N} = \frac{1}{N} \sum_{k=1}^{N} \ln(\frac{X_k}{b}). \tag{6}$$

- Self-similarity: $(n^{-\frac{1}{\alpha}} \Delta S_{t_1^{(n)}}, \ldots, n^{-\frac{1}{\alpha}} \Delta S_{t_k^{(n)}}) \approx (\Delta S_{t_1}, \ldots \Delta S_{t_k}), \Delta t_i^{(n)} = n\Delta t_i = n\Delta.$

Volatility

- Lots of notations and misconceptions: volatility, changeability, measure of uncertainty.
- If $\tilde{r}_k = \mu + \sigma\epsilon_k, \epsilon_k \approx \mathcal{N}(0,1)$, the volatility means the standard deviation σ. 2-σ rule: $\mathbb{P}(\epsilon_k \in (-2,2)) \approx 95\%$. Homoscedasticity!
- In reality, heteroscedasticity - time varying:

$$\mathbb{E}(\tilde{r}_k|\mathcal{F}_{k-1}) = \mu_k,\ \mathbb{E}((\tilde{r}_k - m_k)^2|\mathcal{F}_{k-1}) = \sigma_k^2;$$

$$\langle \tilde{r} \rangle_n = \sum_{k=1}^{n} \sigma_k^2 - \text{quadratic characteristic.} \tag{7}$$

- Engle (Nobel 1982): Autoregressive Conditional Heteroskedasticity - ARCH(p) models

$$\sigma_k^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i \tilde{r}_{k-i}^2. \tag{8}$$

- Nonparametric method: $\tilde{r}_k = \mu_k + \sigma_k\epsilon_k, \hat{\sigma}_k^2 = \frac{1}{k-1}\sum_{i=1}^{k}(\tilde{r}_i - \bar{r}_k)^2$.

Volatility

- Black-Scholes' option pricing formula: Assume $S_t = S_0 e^{(\mu - \frac{\sigma^2}{2})t + \sigma W_t}$.

$$\begin{aligned} C_T &= \mathbb{E}(S_T - K)^+ = (S_0 - K)\Phi(\frac{S_0 - K}{\sigma\sqrt{T}}) + \sigma\sqrt{T}\varphi(\frac{S_0 - K}{\sigma\sqrt{T}}); \\ \varphi(x) &= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \Phi(x) = \int_{-\infty}^{x} \varphi(y)dy. \end{aligned} \tag{9}$$

  Then $C_t(\sigma, T) = e^{-\rho(T-t)} C_T$ is the theoretical call option price.
- In practice $\hat{C}_t = C_t(\tilde{\sigma}_t, T)$ is used to solve $\tilde{\sigma}_t$ - the implied volatility.
- Δ-volatility of order $\delta > 0$: $H_t = \ln S_t$ with

$$\nu^{(\delta)}_{(a,b]}(H, \Delta) = \frac{Var^{(\delta)}_{(a,b]}(H, \Delta)}{[\frac{b-a}{\Delta}]}, \ Var^{(\delta)}_{(a,b]}(H, \Delta) = \sum_{t_k \in (a,b]} |H_{t_k} - H_{t_{k-1}}|^{\delta}. \tag{10}$$

Correlation and cluster

- $|\tilde{r}_k| = \nu_{(t_{k-1},t_k]}(H, \Delta)$ then

$$\rho(k) = \frac{\mathbb{E}\tilde{r}_n\tilde{r}_{n+k} - \mathbb{E}\tilde{r}_n\mathbb{E}\tilde{r}_{n+k}}{\sqrt{D\tilde{r}_n D\tilde{r}_{n+k}}}; R(k) = \frac{\mathbb{E}|\tilde{r}_n||\tilde{r}_{n+k}| - \mathbb{E}|\tilde{r}_n|\mathbb{E}|\tilde{r}_{n+k}|}{\sqrt{D|\tilde{r}_n|D|\tilde{r}_{n+k}|}} \tag{11}$$

- The autocorrelation $\hat{R}(\theta)$ is large for small θ but decreases slowly as θ grows. For monthly period: $\hat{R}(\theta) \approx k\theta^{-\alpha}$ as $\theta \to \infty$ rather than $ke^{-\theta^\beta}$.
- Cluster property (Mandelbrot): large values of volatility are followed by large values, and small values are followed by small values.

Rescale range test

- R/S analysis for long memory/persistence effect: $\mathcal{Q}_n = \frac{\mathcal{R}_n}{\mathcal{S}_n}$ where

$$\mathcal{R}_n = \max_{k\leq n}(H_k - \frac{k}{n}H_n) - \min_{k\leq n}(H_k - \frac{k}{n}H_n),$$
$$\mathcal{S}_n^2 = \frac{1}{n}\sum_{k=1}^{n}(r_k - \bar{r}_n)^2.$$

  Then $\mathcal{Q}_n$ is invariant under linear transformation $r_k \mapsto c(r_k + m)$.
- Donsker-Prokhorov invariant principle: If $r_k \approx \mathcal{N}(0,1)$ are i.i.d. with $B_t^0 = B_t - tB_1$ - the Brownian bridge

$$\mathcal{Q}_n \approx R^* \sqrt{n}, \text{ as } n \to \infty, \text{ where } R^* = \sup_{t\leq 1} B_t^0 - \inf_{t\leq 1} B_t^0 \qquad (12)$$

- Hurst (1951) found with Nile river data that $\mathcal{Q}_n \approx Cn^H$ for some $H > \frac{1}{2}$. Theoretically, it is proved for fractional Brownian motions B_t^H.

Portfolio selection theory

- Markowitz's portfolio diversification
- Capital asset pricing model (CAPM)
- Arbitrage pricing theory (APT)

MPT

- Before, investing processes were centered on individual stocks; investors find assets that would produce decent returns without subjecting the investor to too much risk. Expected net present value (NPV) was used to distinguish these "sure bet" stocks, securities were valued by future DCF. Stocks with revenue and profit growth at a quicker rate were given great value.
- Markowitz (1952) found a statistical approach to support diversification idea believed by economists. Mean and variance should be used to form a diverse portfolio.
- MPT was based on two main concepts: 1, an investor aims to maximize return for any level of risk. 2, risk can be reduced by diversifying a portfolio through individual, unrelated securities.
- MPT's assumption: investors are risk-averse, preferring less risk for a given level of return. Thus, they will only take on high-risk investments if they can expect a larger reward. Unsystematic risk can be reduced.

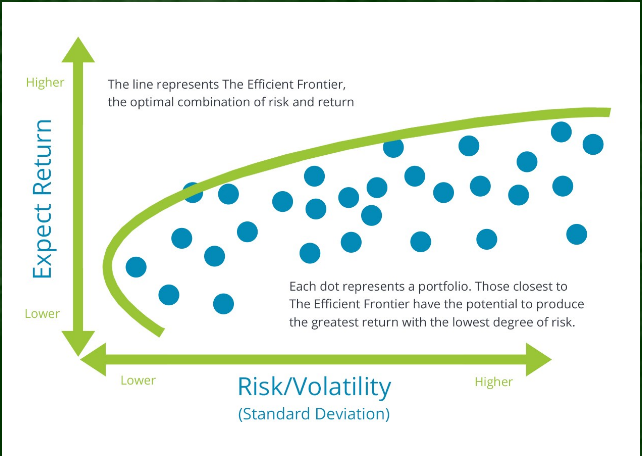
Efficient frontier

- Assumptions: 1, the markets are efficient; 2, the assets follow a normal distribution; 3, markets absorb information quickly and accordingly base the actions; 4, the decisions of the investors are always based on expected return and standard deviation as a measure of risk.
- Minimize risk with constrains:

$$\min\{\omega^T \Sigma \omega \quad | \quad \omega \in R^n, 1^T \omega = 1, \mu^T \omega = \alpha\} \tag{13}$$

- Lagrangian multipliers:
  $L(\omega, \lambda_1, \lambda_2) = L(\bar{\omega}) = \omega^T \Sigma \omega + \lambda_1 (1^T \omega - 1) + \lambda_2 (\mu^T \omega - \alpha)$. Solving the equation $\frac{\partial L}{\partial \bar{\omega}} = 0$ gives the results.
- Exercise!
- Key factor 1: volatility as a measure of risk. But market often get low volatility at peak, and high volatility at bottom!
- Key factor 2: covariance matrix. Is it autonomous? What happens when market goes up or down?

Efficient frontier



CAPM

- Sharpe (1964) significantly advanced the Efficient Frontier and Capital Market Line concepts in his derivation of the CAPM. Lintner (1965) derived the CAPM from the perspective of a corporation issuing shares of stock. Finally, in 1966, Mossin also independently derived the CAPM, explicitly specifying quadratic utility functions.
- Assumptions: (1) EMH; (2) rational, risk-averse investors.
- Cost of equity: security market line (SML)

$$ER_s = R_f + \text{risk premium} = R_f + \frac{cov(R_s, R_m)}{var(R_m)}(ER_m - R_f), \qquad (14)$$

  where R_f- risk free rate, R_m-market return. Implication: low volatility results in low cost of equity!
- Critics: empirical data can not be used to estimate future return! β also unstable in time. Also, R_f - T-bill bond rate (1 years) is time varying.

Sharpe ratio

- Sharpe ratio: $Sh_a = \frac{E(R_a - R_f)}{\sqrt{Var(R_a - R_f)}}$. If $R_f \equiv constant$, then $Var(R_a - R_f) = Var(R_a)$, thus this ratio measures the reward/risk ratio.
- The higher an asset's Sharpe ratio, the better its return has been relative to the risk it has taken on. Greater than 2 is good, while bigger than 3 is excellent.
- Risk is associated with the volatility (variance) of the stock prices.

APT

- APT suggests that within an equilibrium market, rational investors will implement arbitrage such that the equilibrium price is eventually realised. As such, the expected return of an asset is a linear function of various factors or theoretical market indices, where sensitivities of each factor is represented by a factor-specific beta coefficient or factor loading. The linear factor model structure of the APT is used as the basis for evaluating asset allocation, the performance of managed funds as well as the calculation of cost of capital.
- risky asset returns are said to follow a factor intensity structure if they can be expressed as:

Stochastic interest rates

- Bond price: $B_t = B_0 \exp\{\int_0^t r_s ds\}$, where it assumes that

$$dr_t = a(t, r_t)dt + b(t, r_t)dW_t. \tag{15}$$

- Merton model (1973): $dr_t = \alpha dt + \gamma dW_t$.
- Vasicek model (1977): $dr_t = (\alpha - \beta r_t)dt + \gamma dW_t$.
- Dothan model (1978): $dr_t = \alpha r_t dt + \gamma r_t dW_t$.
- Cox-Ingersoll-Ross (CIR) model: $dr_t = k(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$.

Volatility model

One considers the time varying model: $dS_t = \mu_t S_t dt + \sqrt{\nu_t} S_t dB_t$ where

- Vasicek model: $d\nu_t = (\alpha - \beta\nu_t)dt + \gamma dW_t$
- Heston model: $d\nu_t = k(\theta - \nu_t)dt + \sigma\sqrt{\nu_t} dW_t,\ cov(B, W) = \rho$.
- GARCH model: $d\nu_t = k(\theta - \nu_t)dt + \sigma\nu_t dW_t$.
- Non-parametric with un-regular data:

$$\hat{\sigma}^2 = \Big(\frac{1}{n-1}\sum_{k=1}^{n} \frac{(H_{\tau_{k+1}} - H_{\tau_k})^2}{\tau_{k+1} - \tau_k}\Big) - \frac{1}{n}\frac{(H_{\tau_n} - H_{\tau_0})^2}{\tau_n - \tau_0} \qquad (16)$$

Stock price model

- Samuelson's model: $dS_t = S_t(\mu dt + \sigma dW_t)$. Solution

$$S_t = S_0 \exp\{(\mu - \frac{\sigma^2}{2})t + \sigma W_t\}. \tag{17}$$

- Geometric rough path model: $dS_t = S_t(\mu dt + \sigma dX_t)$. How to solve?

$$S_t = S_0 \exp\{\mu t + \sigma X_t - \frac{\sigma^2}{2}[X, 2]_t\}. \tag{18}$$

Importantly, $[X, 2]$ might be a stochastic process.

Risk measures

- Definition of risk. Difference between risk (quantifiable) and uncertainty (unmeasurable).
- Level of risk: policy risk, credit risk, macro risk, market risk.
- Types of risk: volatility, value at risk (VAR), expected shortfall (ES),...